

COSC2671: Social Media and Network Analysis

Assignment 2: Team Worksheet

Group 15 | Postgraduate | RMIT University | Semester 1, 2026
Project: Do Financial Influencers Move the Market?
A YouTube and ASX Social Network Analysis

Member	Student Number	Contribution	Role
Abhishek Mathur	s4200870	33%	Team Lead: Data Collection and Network Analysis
Meishva	s4150918	33%	Sentiment Analysis and Topic Modelling
Ananya	s4145593	33%	Community Detection, Homophily and Report Writing

Section 1: Team Project Plan

1.1 Project Overview

This project explored whether financial influencers on YouTube have the ability to predict or influence movements in the Australian Securities Exchange (ASX). To investigate this, the team analysed 387 videos and 15,811 comments from 53 YouTube channels, covering the period from January 2024 to December 2025. This was combined with daily ASX price data for 10 selected stocks. A range of techniques from COSC2671 were applied, including sentiment analysis, topic modelling, social network analysis, community detection, information diffusion, and homophily testing.

1.2 Problem Formulation and Research Questions

The research question was finalised during a team meeting in Week 9. Several potential topics were initially considered, such as cryptocurrency sentiment on Reddit and political discussions on Twitter. However, YouTube financial influencers and the ASX were ultimately chosen because the topic is unique, feasible using available APIs, and highly relevant to the Australian financial landscape. The study was guided by four key research questions:

- RQ1: Does the sentiment expressed in financial influencer video titles correlate with subsequent ASX returns?
- RQ2: What distinct topic communities exist within the YouTube finance ecosystem, and do they differ in ASX performance?
- RQ3: What is the network structure, and which channels occupy the most influential network positions?
- RQ4: Do audiences exhibit homophily, and how does information about key ASX events spread across the network?

1.3 Phase-by-Phase Project Plan

Phase	Tasks	Owner(s)	Week	Output
1: Setup and Planning	Research question finalisation; API key setup; GitHub repository creation; literature review on social media and market prediction; assignment specification review	All	Wk 9	GitHub repo; API credentials; confirmed research questions
2: Data Collection	YouTube Data API v3 integration; 20 search queries for ASX finance content; video metadata, comment collection (200/video); yfinance ASX download (10 tickers); raw JSON and CSV export	Abhishek	Wk 9	raw_youtube_data.json; videos_clean.csv; comments_clean.csv; asx_prices.csv
3: Preprocessing	Emoji removal; English-language detection (70% ASCII); duplicate removal; domain stopword list (70 terms); video text cleaning; ASX	Meishva	Wk 9–10	videos_filtered.csv; comments_filtered.csv; asx_prices.csv (verified)

	T+0/T+1/T+3 date alignment; yfinance column fix			
4: Sentiment Analysis	VADER on video titles and 15,811 comments; channel-level aggregation; Pearson correlation with ASX returns; t-tests; weekly time-series overlay with ASX	Meishva	Wk 10	videos_with_sentiment.csv ; 4 sentiment plots; correlation statistics
5: Topic Modelling	LDA corpus construction (CountVectorizer, 800 features); n=2 to 12 evaluation via log-likelihood and perplexity; 5-topic final model; topic labelling; channel heatmap; ASX return by topic	Ananya	Wk 10	videos_with_topics.csv; topic_summary.csv; 5 topic plots
6: Network Construction	Commenter-overlap graph in NetworkX (37 nodes, 172 edges); degree, betweenness, eigenvector, clustering, PageRank centrality; network stats; GEXF export; 4 visualisation plots	Abhishek	Wk 10	node_attributes.csv; commenter_overlap_network.gexf; 4 network plots
7: Community Detection	Louvain community detection (modularity 0.394); 7 communities profiled by topic, sentiment, ASX returns; inter-community edge analysis; one-way ANOVA	Ananya	Wk 11	community_assignments.csv; community_summary.csv; 4 community plots
8: Diffusion and Homophily	First-mover analysis for 8 key ASX topics; cross-correlation at lags -3 to +3 weeks; sentiment, topic, and community homophily tests; Trump tariff cascade timeline	Abhishek	Wk 11	diffusion_results.csv; homophily_results.csv; 4 diffusion and homophily plots
9: Report and Presentation	Full 20-page report drafting; all figures and captions; worksheet; self-reflections; presentation slide deck; rehearsal	All	Wk 11	Assignment2_Report.docx; worksheet_PG_Group15.pdf.docx; Presentation slides

Note on version control: All code files were developed and iterated locally on individual machines throughout the project period. The GitHub repository was used as a centralised sharing and final submission tool, with the complete codebase pushed on 24 May 2026 prior to submission. This reflects the team's working practice of local development with a single consolidated push rather than incremental commits.

1.4 Tools and Technologies

Tool / Library	Purpose
Python 3.12 (venv)	Primary implementation language for all analysis

YouTube Data API v3	Video metadata, comment, and channel data collection
yfinance	ASX daily OHLCV price data download (10 tickers, 507 days)
pandas, numpy	Data manipulation, cleaning, and alignment
vaderSentiment	Social media sentiment scoring (VADER lexicon)
scikit-learn (LDA)	Topic modelling with perplexity and log-likelihood evaluation
NetworkX	Graph construction, centrality measures, network statistics
python-louvain	Louvain community detection algorithm
matplotlib, seaborn	All visualisations: plots, heatmaps, network graphs
scipy	Pearson correlation, t-tests, ANOVA, cross-correlation
nltk	Stopwords and WordNet lemmatisation for text preprocessing
GitHub	Version control, code sharing, and progress tracking

1.5 Work Breakdown and Contribution Summary

Member	Primary Responsibilities	Approx. Hours
Abhishek Mathur	Data collection (File 01), network construction (File 05), information diffusion and homophily (File 07), project coordination, final code review	~27 hrs
Meishva	Data preprocessing (File 02), sentiment analysis (File 03), ASX alignment, correlation analysis, literature review, report Sections 5 and 6	~27 hrs
Ananya	Topic modelling (File 04), community detection (File 06), full report writing (all sections), worksheet, presentation slides	~27 hrs

Section 2: Weekly Timesheets

2.1 Timesheet: Abhishek Mathur (s4200870)

Week	Date Range	Task Description	Hrs
Week 9	7 May – 10 May 2026	Led team meeting to finalise research question and scope. Set up GitHub repository and branch structure. Registered YouTube Data API v3 credentials. Developed File 01 (01_data_collection.py): implemented 20 search queries, video metadata retrieval, comment collection (200 per video), transcript collection, and yfinance ASX download across 10 tickers. Ran initial collection pass yielding 361 videos and 36,403 comments. Coordinated with team on shared CSV naming conventions.	10
Week 10	11 May - 17 May 2026	Expanded search queries and re-ran data collection to increase dataset to 732 videos and 40,986 comments across 193 channels. Developed File 05 (05_network_construction.py): built commenter-overlap NetworkX graph (37 nodes, 172 edges); computed degree, betweenness, eigenvector, clustering coefficient, and PageRank centrality measures; calculated network statistics (density 0.258, avg clustering 0.605, avg path length 1.77); generated 4 network visualisation plots; exported GEXF for Gephi. Reviewed Meishva's preprocessing output and confirmed ASX alignment was correct for 385 of 387 videos.	10
Week 11	18 May – 24 May 2026	Developed File 07 (07_information_diffusion_homophily.py): implemented first-mover analysis for 8 key ASX topics; conducted cross-correlation between content volume and ASX returns at lags -3 to +3 weeks; performed sentiment homophily ($r=0.120$, $p=0.117$), topic homophily ($t=1.60$, $p=0.112$), and community homophily structural analysis; generated 4 diffusion and homophily plots. Validated findings against real 2024-2025 ASX events. Contributed to final report review, code commenting, and README preparation.	7
TOTAL			27

2.2 Timesheet: Meishva Kiran Rathod (s4150918)

Week	Date Range	Task Description	Hrs
Week 9	7 May – 10 May 2026	Conducted literature review on social media sentiment and financial market prediction (Bollen et al. 2011; Tetlock 2007; Bartholomeusz 2019).	10

		Developed File 02 (02_preprocessing.py): implemented emoji removal using Unicode regex, English-language detection filter (70% ASCII alphabetic threshold), minimum comment length filter (10 characters), duplicate removal, domain-specific stopword list construction (70 terms), video title and description cleaning, and ASX trading day alignment for T+0, T+1, and T+3 return windows. Resolved yfinance multi-level column bug and fixed ASX data download pipeline.	
Week 10	11 May - 17 May 2026	Developed File 03 (03_sentiment_analysis.py): applied VADER sentiment analysis to all 387 video titles and 15,811 cleaned comments; computed channel-level aggregation (avg comment sentiment and title sentiment per channel); ran Pearson correlation between title sentiment and ASX returns at T+1 ($r=-0.005$, $p=0.920$) and T+3 ($r=-0.069$, $p=0.177$); compared mean T+1 returns by sentiment label; ran t-test between bullish and bearish video groups; generated 4 sentiment plots including weekly sentiment timeline overlaid on ASX closing price. Wrote Sections 3 and 5 of the report and the analysis approach table.	10
Week 11	18 May – 24 May 2026	Assisted with report proofreading and verification of all statistical values cited in text. Compiled result tables for Sections 5, 9, and 10. Verified all 20 figure captions against code output files. Contributed to presentation slide content for preprocessing and sentiment analysis sections. Cross-checked references and bibliography formatting.	7
TOTAL			27

2.3 Timesheet: Ananya Shandilya (s4145593)

Week	Date Range	Task Description	Hrs
Week 9	7 May – 10 May 2026	Reviewed LDA topic modelling literature and sklearn documentation. Conducted exploratory preprocessing experiments to understand corpus characteristics. Assisted Meishva with construction of domain-specific finance stopword list. Set up local Python virtual environment and confirmed all package dependencies (nltk, scikit-learn, pyLDAvis, matplotlib) were correctly installed. Drafted initial version of the team project plan and weekly timeline.	8
Week 10	11 May - 17 May 2026	Developed File 04 (04_topic_modelling.py): built CountVectorizer document-term matrix (385 documents, 800 features, min_df=2, max_df=0.90, unigram and bigram range); evaluated $n=2$ to 12	11

		LDA topics via log-likelihood and perplexity; trained final 5-topic model; interpreted and labelled all five topics (ETF & Dividends, Stock Research & Tips, Beginner Guides & Education, Global Markets & Sectors, ASX Blue Chips & News); linked topics to ASX T+1 return data; generated 5 topic visualisation plots. Began drafting Sections 4 and 6 of the final report.	
Week 11	18 May – 24 May 2026	Developed File 06 (06_community_detection.py): ran Louvain community detection achieving modularity 0.394 with 7 communities; profiled Community 0 (Global Finance, 11 channels) and Community 1 (Australian Finance, 21 channels); computed inter-community shared commenter matrix (84 cross-community commenters); ran one-way ANOVA ($F=0.60$, $p=0.61$); generated 4 community plots. Wrote and compiled full 20-page assignment report (all 11 sections). Produced team worksheet (PDF and Word formats). Prepared and formatted all presentation slides.	8
		TOTAL	27

Section 3: Individual Self-Reflections

Abhishek Mathur (s4200870)

It was quite challenging but rewarding to take on the same assignment technically; it would also give me the chance to use network analysis in a real problem situation concerning financial social media. The major contribution I made was to create the data collection and network building pipelines. I discovered while working with the YouTube API that quota limits, repeated searches, and reproducible collection were more complex than I'd anticipated. The most intriguing aspect of my work was the creation of the 'commenter overlap network', particularly because Bryan Invest was ranked highest in terms of the following: degree, betweenness, eigenvector centrality, and PageRank. This facilitated a robust structural interpretation of influence in the network. Worked well from a team perspective, with CSV naming conventions and GitHub making it a bit easier to integrate. I would try to set up more frequent mid-week check-ins, collect the transcripts while they are still available for more depth in our NLP analysis, and in future projects would try to have more frequent check-ins.

Meishva (s4150918)

I was able to learn about the challenges of using NLP with real data in social media through this assignment. The primary role given to me was to use data preprocessing and draw out sentiments. The comment system was quite messy, contained multiple languages, included emojis, and was often very short, making it more complicated than anticipated to do a clean up. I did some work on non-english comments filtering, duplication removal, text cleaning and date syncing of videos with the ASX return windows. The use of VADER sentiment analysis on titles and comments was informative as it demonstrated a clear difference in the framing of the influencers, and the reactions from the audience. What was most valuable for me to learn is that a null result can be meaningful: When I analyzed the relationship between video title sentiment and next-day ASX returns, there was virtually none, with a correlation coefficient of $r = -0.005$. In the future, I would come up with more specific search queries related to specific financial events; otherwise, I missed some important materials regarding tariffs.

Ananya (s4145593)

This project provided me with great experience in using the topic modelling, community detection, and report writing on a real world social media data set. My main contributions were the topic modeling with LDA, the community detection with Louvain method, providing support for the homophily analysis and preparing the final report. Creating an appropriate stopwords list was the most difficult part of topic modelling; initial topic models yielded noisy topics, but after a few iterations the five final topics were interpretable. The interesting part of the community detection results was that global finance channels were clearly separated from Australian finance channels with a modularity score of 0.394. I also gained an understanding of the value of cohering technical findings to the research questions. For future group work I would give the report writing tasks earlier in the group for less pressure at the end of the week; however, the one main editor helped to keep the writing consistent.

